



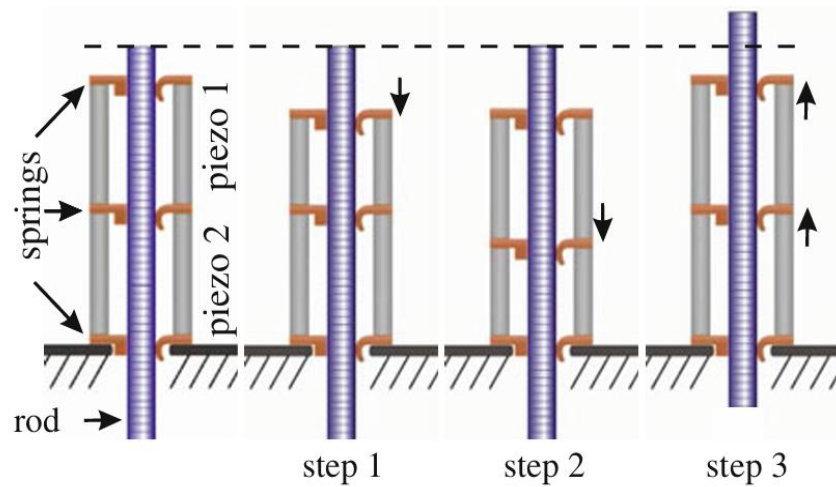
Building KoalaDrive Piezomotor Type 2 for STM

Scanning Tunneling Microscopy & Transport Lab

Final Presentation

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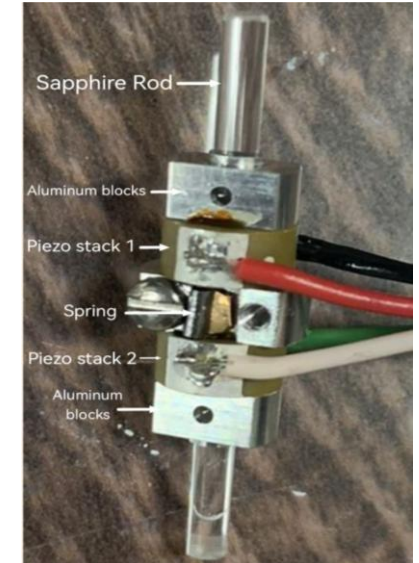
Original KoalaDrive

Limitation:

- High threshold voltage
- Device is compact, but further miniaturization not explored

Developing a more efficient KoalaDrive

Change the design



KoalaDrive Type 2

Comparison	Original KD	KD Type 2
Size	20 mm	16 mm
Threshold Voltage	± 30 V	± 15 V
Friction Mechanism	Spring	Sapphire ball
Spring Type	Commercial	Homemade (tuneable)
Rod	Stainless steel	Sapphire

Methods



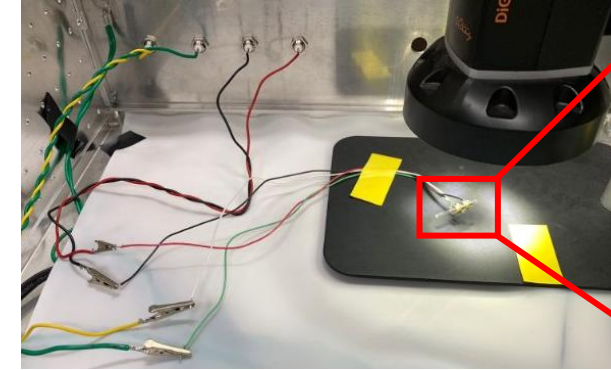
Non-conductive epoxy
(H74F)



Cured epoxy
on a hotplate



High Voltage Amplifier



Circuit testing setup



Vertical and
Horizontal
testing

Attach
sapphire balls
+ aluminum
block

Attach the
sapphire ball
+ copper foil

Formed
spring

Assembled
full body of
KD

Wiring

Testing

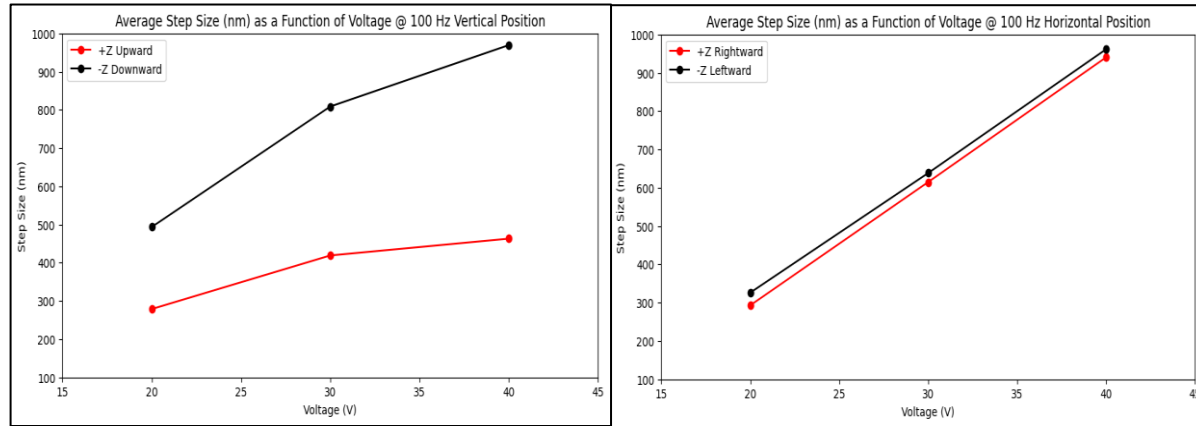
Control Parameters:

- **Voltage**
- **Frequency**
- **Alignment** (horizontal & vertical)

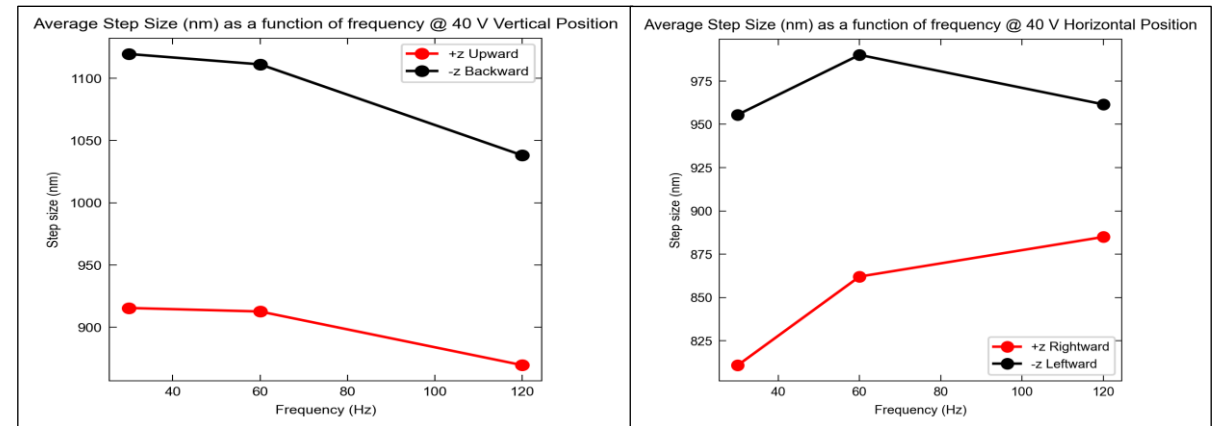
- Successfully tested at **Room Temperature (RT)**
- Functional tests were attempted at Liquid Nitrogen (LN2) temperature, but results were inaccurate
- No ultra-high vacuum (UHV) testing due to time constraints

Data Processing and Comparison

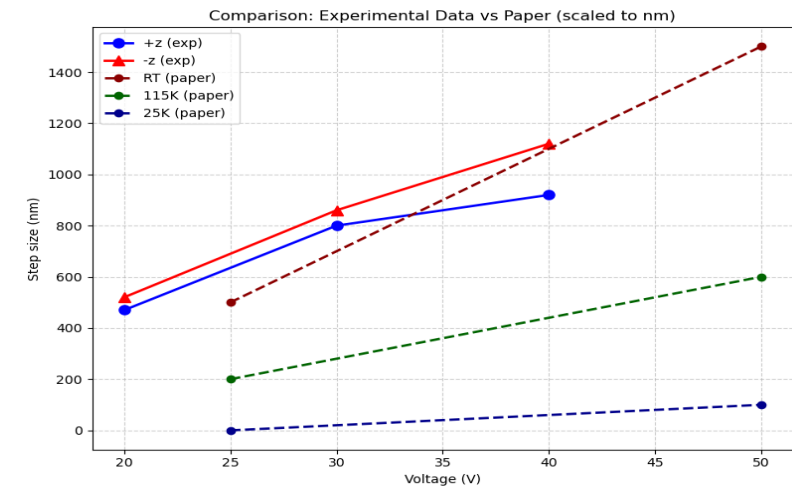
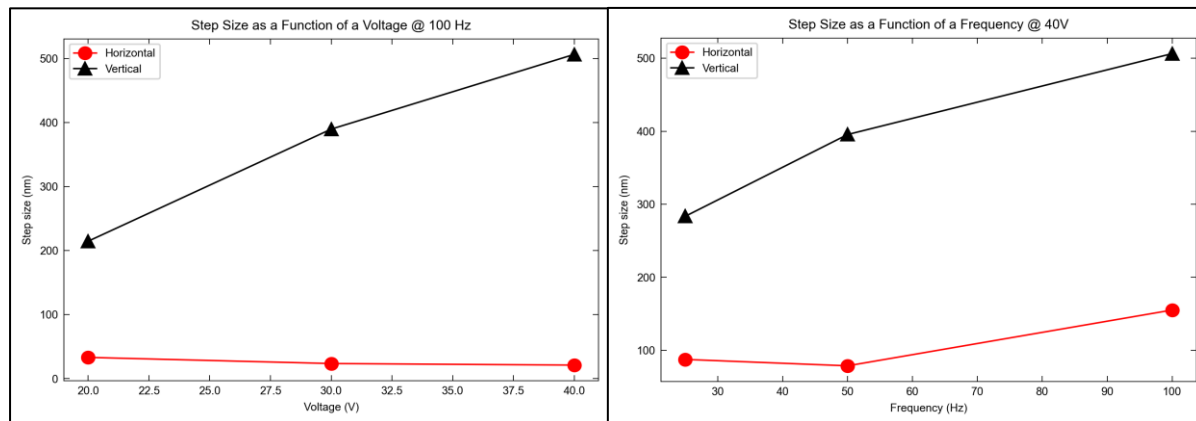
1. Control Voltage



2. Control Frequency

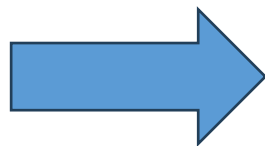
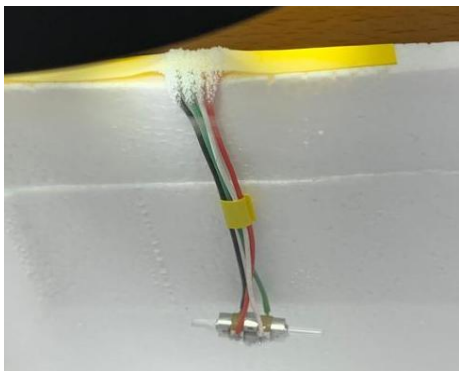


3. Control Alignment



Compare data from Cherepanov et al., Review of Scientific Instruments (2012) with experimental data

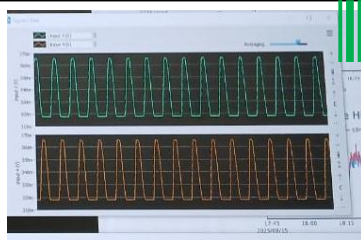
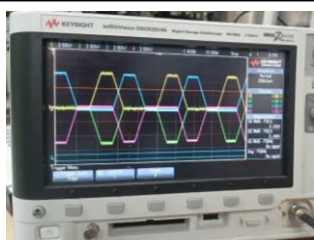
Temperature Control



Upper Piezo Part	RT Capacitance: 0,200 F	LN2 Capacitance: 0,055 F	(3,64) factor reduce
Lower Piezo Part	RT Capacitance: 0,190 F	LN2 Capacitance: 0,052	(3,65) factor reduce

- Note:** For low temperature, the results are not accurate. So, UHV is required for testing

Trying to test KD Type 2 in liquid nitrogen (LN2, 77K)



Triangle wave signal from supply

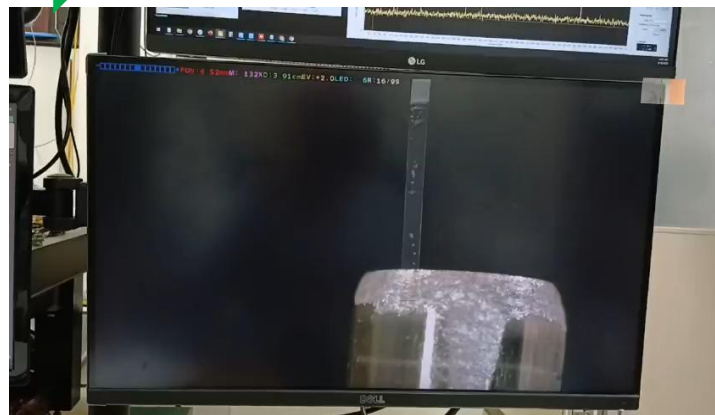
Performance Comparison (Our Experiment vs Cherepanov et al., 2012)

RT:
Our data shows ~1.0–1.2× larger step size compared to RT data in the paper.

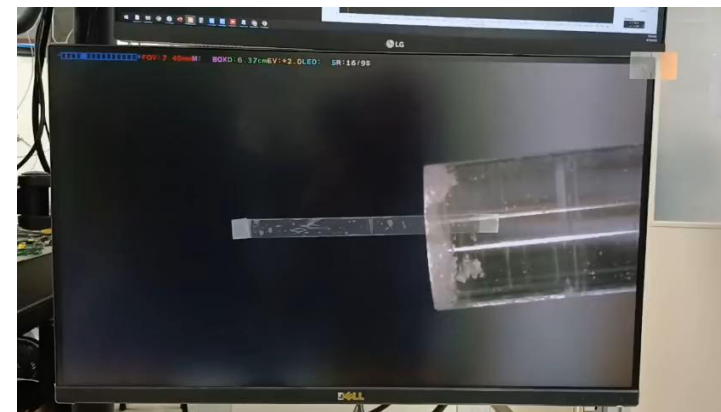
Liquid Nitrogen (115 K):
Our step size is ~2–2.5× higher at the same voltage range.

Helium (25 K):
Our performance reaches ~10–15× larger displacement.

Result:



Piezo movement +Z (upward) and -Z (downward) vertical position video



Piezo movement +Z (rightward) and -Z (leftward) horizontal position video